

Suzanne Lafon

PhD in Physics

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CONTACT

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- **Website** suzannelafon.github.io
- **Github** https://github.com/slafon11
- **Linkedin** Suzanne Lafon
- **ORCID** 0000-0002-3441-2041

DIPLOMA

- **PhD in Physics**, Paris-Saclay University
- **Master of fundamental physics (ICFP)** *"Soft matter and physics for biology"*, ENS Ulm, Paris
- **ENS diploma**
  4 years study in Physics (major) and 72 additional ECTS in other domains
- **Bachelor in fundamental physics**
  ENS Ulm, Paris
- **Classe Préparatoire (CPGE) PCSI-PC***
  Lycée Joffre, Montpellier

AWARDS

- **PhD thesis award** from the Paris-Saclay Graduate School of Physics

SKILLS

Computing

Python, Matlab, LAMMPS

Experimental

- *Scattering & Reflectivity techniques (X-rays and neutrons)*
- *Microfluidics (on glass & PDMS)*
- *Surface chemistry*
- *Rheology*
- *STRAFI NMR*

WORK EXPERIENCE

Mar 2027 – Oct 2025
Postdoc
LIPHY, CNRS · Grenoble, France
Quasi-elastic study of ionic liquids confined in a sheared polymer matrix
with M. Plazanet (CNRS)

Sep 2025 – Oct 2024
Postdoc
LIPHY, CNRS · Grenoble, France
Chiral molecules separation by chiral cellulose derivatives - a MD study
with B. Coasne (CNRS)

Sep 2024 – Juin 2024
Postdoc
Paris-Saclay University · Orsay, France
Ion dynamics: development of a STRAFI-NMR set-up
with M. Zeghal (Paris-Saclay University), P. Judeinstein (CNRS) and P. Wzietek (Paris-Saclay University).

Nov 2023 – Oct 2020
PhD thesis
Paris-Saclay University · Orsay, France
Founded by ENS Paris.
Liquid/solid interfacial rheology
supervised by F. Restagno (CNRS) and A. Chennevière (CEA Saclay) in collaboration with L. Joly and S. Merabia (ILM, Lyon, France).

Nov 2023 – Oct 2020
Teaching assistant
Paris-Saclay University · Orsay, France
Computer science (Python, L1), metrology (IUT S4) et material science (L3).

2023 – 2021
Non-permanents representative
Solid State Physics Laboratory · Orsay, France

2021 – 2020
Organiser of the "Rencontre des Jeunes Physicien·ne·s"
French Physical Society (SFP) · Paris, France

Sep 2020 – Sep 2016
Research internships
Multiples institutions · France & Allemagne
Spontaneous emulsification and spontaneous organisation at the oil/water interface, Gulliver Lab, ESPCI, Paris, France. Supervised by Teresa Lopez-Leon. 5 months.
Photoresponsive liquid/liquid phase separation, CRPP, Pessac, France. Supervised by Nicolas Martin & Jean-Christophe Barret. 5 months.
Topological defects in cellulose shells, Gulliver lab, ESPCI, Paris. Supervised by Teresa Lopez-Leon. 4 months.
Effect of advection on pulse propagation in excitable media, MPIDS, Göttingen, Germany. Supervised by Jean-Daniel Julien & Karen Alim. 5 months.
Short-times dynamics of capillary rise, PMMH, ESPCI, Paris. Supervised by Joachim Delannoy, Etienne Reyssat & David Quéré. 1 month.

2017
Head of the Sports Associaion
Ecole Normale Superieure de Paris · Paris, France

FORMATIONS

2024	**Machine learning with Python** *MOOC of INRIA*
2023	**Image processing (Python)** *F. Boulogne* *(Paris-Saclay Univ.)*
2022	**Graphical interfaces (Python)** *G. Exil* *(CEA Saclay)*
2021	**Methodology for an open and reproducible research** *MOOC by C. Pouzat,* *A. Legrand and K. Hinsen (INRIA)*
2021	**History of chemistry** *J.-B. Baudin at ENS Ulm, Paris*

LANGUAGES

French	Mother tongue
English	Fluent
Spanish	Intermediate

CONTACTS

● **Postdoc Research Advisors**

Benoit Coasne
benoit.coasne@univ-grenoble-alpes.fr

Marie Plazanet
marie.plazanet@univ-grenoble-alpes.fr

Patrick Judeinstein
patrick.judeinstein@cnrs.fr

● **PhD Supervisors**

Frédéric Restagno
frederic.restagno@universite-paris-saclay.fr

Alexis Chennevière
alexis.chenneviere@cea.fr

PUBLICATIONS

2025 — **Chiral separataion by chiral structures**
IN PREPARATION
S. Lafon, S. Nono-Tagne, M. Plazanet, Y. Nishiyama, Y. Ogawa, I. Ostuka, B. Coasne

2025 — **NMR study of water diffusion in highly concentrated salt**
IN PREPARATION
S. Lafon, J. Vedel, C. Teynier, D. Raj-Mithalal, A. Kayi, M. Zeghal, P. Wzietek, P. Judeinstein

2025 — **A new method to quantify liquid/solid friction using a rheometer**
IN PREPARATION
S. Lafon, D. Dudzinski, F. Restagno, A. Chennevière

2025 — **Near-Surface Concentration Profile of Sheared Semidilute Polymer Solutions**
Langmuir, **41**(3) & arxiv
S. Lafon, T. Outerelo Corvo, M. Grzelka, A. Hélary, P. Gutfreund, L. Léger, A. Chennevière, F. Restagno

2025 — **Probing the colloidal behavior of a cell wall polysaccharides-degrading enzyme in a highly constrained model system**
J. Coll. Int. Sci., **694**(137685)
H. Voisin, E. Bonnin, M. Marquis, C. Alvarado, **S. Lafon**, T. Lopez-Leon, F. Jamme, I. Capron

2023 — **Giant slip length at a supercooled liquid-solid interface**
PRE, **107**(2) & arxiv
S. Lafon, A. Chennevière, F. Restagno, S. Merabia, L. Joly

2023 — **Flow driven control of pulse width in excitable media**
PRE, **107**(5) & arxiv
A. Misselwitz, **S. Lafon**, J.-D. Julien, K. Alim

2023 — **Interactions between proteins and cellulose in a liquid crystalline media: Design of a droplet based experimental platform**
Int. J. Bio. Macrmol., **245**(125488)
H. Voisin, E. Bonnin, M. Marquis, C. Alvarado, **S. Lafon**, T. Lopez-Leon, F. Jamme, I. Capron

2022 — **Reversible photocontrol of DNA coacervation**
*Methods in Enzymology, **646** (329-351)*
S. Lafon, N. Martin

2019 — **The dual role of viscosity in capillary rise**
*Soft Matter, **15**(13)*
J. Delannoy, **S. Lafon**, Y. Koga, E. Reyssat, D. Quéré

CONFERENCES

Jun 2025 *Talk*	**Journées de la Diffusion Neutronique** Annecy, France
Jul 2023 *Talk*	**International Congress on Rheology** Athènes, Greece
Oct 2022 *Talk*	**Journées de la Diffusion Neutronique** Biarritz, France
Jul 2022 *Poster*	**Soft Matter Summer School** Cargèse, France
Jul 2022 *Poster*	**New frontiers in liquid matter - CECAM workshop** Paris, France
Sep 2021 *Poster*	**ECIS** Athens, Greece
Nov 2019 *Poster*	**Symposium "Synthetic and Systems Biology"** Bordeaux, France